

Solution Requirements

Date	1 July 2026
Team ID	—
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Data Ingestion	System accepts 10 weather/terrain readings (rainfall, humidity, river level, etc.) via a web form	High
2	Preprocessing	Incoming values are scaled with the same StandardScaler used during training before prediction	High
3	Prediction	Trained classifier (best of Decision Tree / Random Forest / KNN / XGBoost) returns a Flood / No-Flood label	High
4	Probability Output	System returns the predicted flood probability (%) alongside the label when the model supports it	Medium
5	Result Presentation	Distinct result pages (flood_result.html / no_flood_result.html) render outcome with plain-language guidance	High
6	Model Transparency	Home page displays which model is active and its test accuracy (from metadata.json)	Low
7	Error Handling	Invalid or missing form inputs redisplay the form with a clear error message instead of crashing	Medium
8	Reproducible Training	Notebook scripts allow retraining the model from a CSV dataset and regenerating artifacts	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should return almost instantly since the model is a lightweight pre-trained pickle	Response < 1 second on local server
2	Scalability	App should handle multiple sequential prediction requests without reloading the model each time	Model/scaler loaded once at startup
3	Security & Data Privacy	No personal data is collected; only anonymous weather/terrain values are submitted	No PII stored or logged

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
4	Reliability & Availability	App should run locally without crashing on malformed input	Graceful fallback to form + error message
5	Usability & Accessibility	Form should use clear labels, sensible min/max ranges and default values for every feature	All 10 fields have labels, units and defaults
6	Maintainability	Feature list and model can be swapped by regenerating metadata.json without code changes	New model drop-in via /model folder